



To: Ron [Last Name], Program Director, Aerospace Classes
From: Kyle Nessen
Date: February 2026
Subject: **Proposed Certificate Structure for UAS Program**

Overview

This document proposes a set of stackable certificates for Cuesta College's emerging UAS (Unmanned Aircraft Systems) program. The structure uses a common foundation course, AERO 100: Basic UAV Operator, as the entry point for all certificate tracks. From that foundation, students select a specialization aligned with their career interests, earning a Certificate of Specialization. Students who complete multiple specializations or a broader set of coursework earn a Certificate of Achievement, the program's primary workforce credential.

This design follows proven institutional models already operating successfully at Cuesta. The Child Development and Family Studies program uses a stacking approach where each certificate level builds on the last: Associate Teacher (12 units) feeds into Teacher (26 units), which feeds into Master Teacher (35 units). The Business Administration program uses a different but complementary pattern: a shared Career Track Core certificate that branches into concentration-specific credentials in Accounting, Marketing, Small Business Management, and other fields. The proposed UAS structure combines both approaches – a common core branching into specializations, with a stacking pathway from smaller credentials to larger ones.

The interdisciplinary nature of drone operations is a strength. By pairing AERO courses with existing offerings in Geography, Agriculture, Art, and Film/Television, the program creates cross-enrollment pathways that serve students from multiple disciplines while strengthening the enrollment base for all participating departments.

Foundation Course

All certificates begin with the approved foundational course:

Course	Title	Units
AERO 100	Basic UAV Operator	3.0

AERO 100 prepares students for FAA Part 107 certification and covers regulations, airspace, weather, flight planning, safety, and hands-on flight proficiency to NIST Level 2 standards. This course is already approved through CurriQunet and serves as the prerequisite for all applied AERO courses.

Proposed New AERO Courses

Four new courses are proposed for the UAS program. Each subsection describes the course purpose, what is needed to bring it to offering, and open questions for discussion. AERO 110 and AERO 120 have Course Outline Reports drafted and ready for review. The two agricultural courses are earlier in development and will require additional collaboration before formal proposals are written.

Course	Title	Units	COR Status
AERO 110	Drone Mapping and Photogrammetry	2.0	Drafted
AERO 120	Thermal Imaging and Applied Inspection	3.0	Drafted
AERO 1XX	Multispectral Imaging and Crop Monitoring	2.0–3.0	Not started
AERO 1XX	Agricultural Spray Operations	2.0–3.0	Not started

AERO 110: Drone Mapping and Photogrammetry

Purpose

Drone mapping and photogrammetry represent the largest commercial drone employment sector. This course teaches students to plan and execute professional mapping missions, from RTK positioning and ground control point methodology through photogrammetry processing and deliverable production. Students produce orthomosaics, digital elevation models, and volumetric calculations meeting professional specifications. The course prepares graduates for positions in surveying firms, construction companies, environmental consulting, civil engineering, and county planning departments.

This is the course closest to ready. The instructor has extensive professional experience in drone mapping and photogrammetry, the curriculum content is well developed, and the core equipment is in hand.

What It Takes to Offer

Equipment on hand:

- DJI Matrice 4E mapping drone (secured through separate funding)
- DJI training fleet (Mini 3s and Air 3s for AERO 100, which build prerequisite flight skills)

Equipment needed:

- Emlid Reach RS4 base station (\$3,900) and Reach RX2 rover (\$2,400) for RTK positioning
- Base and rover accessories kit including tripod, tribrach, survey pole, and bipod (\$1,294)
- Ground control point targets and survey markers
- Additional Matrice 4E batteries and field charging infrastructure for extended mapping sessions

The Emlid GNSS equipment is recommended for immediate purchase, as noted in the December 2025 equipment memo. Total RTK investment is approximately \$7,594 before tax. This equipment also supports professional field work and course material development ahead of the first offering.

Facilities needed:

- Computer lab with workstations capable of running photogrammetry processing software. Metashape requires a dedicated GPU, 32+ GB of RAM, and substantial storage. The specific requirements depend on project size, but processing a typical student dataset takes 30–60 minutes on an adequate machine.
- Open outdoor areas with varied terrain for mapping exercises. San Luis Obispo County's topography is an asset here – terrain-following flight planning is a critical skill, and our local landscape provides ideal teaching conditions.

Software needed:

- Agisoft Metashape (desktop photogrammetry processing). Educational licensing should be explored – Agisoft offers educational pricing.
- ArcGIS Pro or QGIS for viewing, analyzing, and exporting spatial data. ArcGIS is the industry standard; QGIS is free and open source. Both are worth teaching.
- DroneDeploy or similar cloud-based processing platform, which allows students to compare cloud versus desktop workflows.
- Emlid Flow (RTK survey operations on mobile devices). Free with Emlid hardware.

Departmental collaboration:

- Geography/Geology department for GEOG 230 alignment. GEOG 230 is already listed as an advisory on the AERO 110 COR. The GIS faculty should review the mapping curriculum to ensure complementary rather than duplicative content, particularly around coordinate systems and spatial data formats.

Open Questions

1. Unit count: 2.0 or 3.0? The drafted COR uses 2.0 units (1 hr lecture / 4 hr lab) based on Ron's suggestion, but the curriculum content is extensive. At 3.0 units (2 hr lecture / 3 hr lab, matching AERO 100), the course has more room for the processing and analysis side. At 2.0, the scope would need tightening, likely assuming students bring GIS background from GEOG 230.

2. Computer lab availability: Is there an existing lab with machines that meet Metashape's requirements? If not, what is the pathway to securing one – new machines in an existing lab, shared access with another department, or cloud-based processing as a workaround?
3. Software licensing: What educational pricing is available for Metashape and ArcGIS Pro? Esri has institutional agreements with many community colleges. Has Cuesta's GIS program already secured ArcGIS licensing that AERO students could access?
4. Course number: Is AERO 110 confirmed, or does a different number make more sense in the department's scheme?

AERO 120: Thermal Imaging and Applied Inspection

Purpose

Thermal imaging from drones serves two major employment sectors: infrastructure inspection and public safety. This course covers infrared theory, thermal camera operation, image interpretation, and professional reporting for both sectors. Students learn to identify thermal anomalies in electrical systems, building envelopes, and solar installations, and to apply thermal imaging to search and rescue, fire assessment, and law enforcement support operations. The course aligns with Level 1 thermography certification requirements, an industry credential that significantly enhances employment prospects.

PG&E represents the single largest local employer of drone operators, with 80+ pilots deployed daily and over 300,000 inspections completed in 2024. The combination of Part 107 certification (from AERO 100) and thermography certification positions graduates competitively for utility inspection positions. Public safety applications serve a different audience – primarily working first responders adding drone skills – and represent a continuing education market.

This course requires more instructor development than AERO 110. The instructor has foundational thermal imaging knowledge but less direct professional experience in this area compared to mapping. Curriculum development will benefit from partnership with industry practitioners, particularly from the utility inspection and fire service sectors.

What It Takes to Offer

Equipment on hand:

- DJI Matrice 4T with thermal payload (secured through separate funding). This drone aligns with PG&E's equipment standards, which is a meaningful advantage for students targeting utility inspection careers.

Equipment needed:

- Thermal calibration targets and controlled heat sources for training scenarios
- Laptop or workstation with thermal analysis software

- Additional Matrice 4T batteries for extended field sessions
- Safety equipment for operations near infrastructure (high-visibility gear, communication radios)

The equipment investment for this course is substantially lower than for mapping, since the primary drone is already in hand and the software requirements are less demanding than photogrammetry processing.

Facilities needed:

- Access to buildings and infrastructure for inspection exercises. Campus buildings could serve as initial training sites for building envelope assessment.
- Suitable outdoor areas with controlled access for night operations (SAR exercises). Night flight adds logistical complexity that needs advance planning.
- Indoor space for thermal physics demonstrations and camera configuration exercises.

Software needed:

- DJI Thermal Analysis Tool (free with DJI hardware)
- FLIR Thermal Studio or similar advanced analysis software
- DJI Pilot 2 for flight operations

Partnerships needed:

- PG&E for curriculum review, equipment standards alignment, and potential hiring pipeline development. Their Paso Robles drone dock operations are local.
- CAL FIRE or SLO County Fire for public safety module input, exercise scenarios, and potential guest instruction.
- Local solar installers for panel inspection training sites.
- Insurance industry contacts for documentation requirements and seasonal employment connections.

Instructor development:

- Level 1 thermography certification for the instructor. Training costs \$1,500–\$5,000 and would strengthen both teaching credibility and the program's relationship with certification bodies.
- Ride-along or observation time with utility inspection operations to build firsthand understanding of professional workflows.
- Consultation with public safety agencies on incident command integration and operational protocols.

Open Questions

1. Partnership development: Has initial contact been made with PG&E or local fire departments? These relationships are important for both curriculum input and eventual job placement.

2. Night operations logistics: SAR exercises require nighttime flight. What are the insurance, facility access, and safety considerations for night lab activities on or near campus?
3. Public safety scope: The COR currently combines infrastructure inspection and public safety in one course. Should the balance shift more toward inspection (the stronger employment pathway) with public safety as a supporting module, or is the current balance appropriate?
4. Continuing education scheduling: Working firefighters and law enforcement officers represent a key audience. Should the course be designed for evening and weekend scheduling from the start?
5. Guest instruction: The public safety modules may benefit from guest instructors from fire and law enforcement. What is the process for arranging guest instruction at Cuesta?
6. Thermography certification integration: Could the college partner with Infrastructure Inspection Institute or the Infrared Training Center to offer Level 1 certification as part of or alongside this course? This would add significant value for students.

AERO IXX: Multispectral Imaging and Crop Monitoring

Purpose

This course teaches students to use multispectral drones for precision agriculture: assessing crop health, identifying stress patterns, detecting pest and irrigation issues, and generating prescription maps for variable-rate fertilizer and chemical application. The agricultural drone market is the fastest-growing segment of commercial UAS operations, with some projections suggesting 80% of future drone usage could come from agriculture.

The course borrows foundational mapping concepts from AERO 110 – flight planning, basic orthomosaic production, coordinate systems – but shifts the focus to multispectral sensor operation, vegetation index calculation (NDVI and related indices), and agronomic data interpretation. Ground control points and RTK positioning receive lighter treatment than in AERO 110, since the accuracy requirements for crop monitoring are less stringent than for survey-grade mapping. Color orthomosaic production is covered as supporting context rather than as the primary deliverable. The emphasis is on getting students efficiently to the point where they can fly a multispectral mission, process the data, interpret the results, and communicate findings to a grower.

What It Takes to Offer

Equipment needed:

- Multispectral-capable drone. The DJI Mavic 3 Multispectral (approximately \$6,500) is purpose-built for agricultural monitoring, with an integrated multispectral camera capturing red, green, red edge, and near-infrared bands alongside

a standard RGB camera. This is a separate purchase from the existing fleet – the Matrice 4T's thermal sensor captures different wavelengths than a multispectral sensor and cannot substitute for it.

- Batteries and accessories for field operations.

Software needed:

- Processing software capable of generating vegetation indices from multispectral data. DroneDeploy, Pix4DFields, or Metashape can all handle this, though the agricultural-specific platforms (DroneDeploy, Pix4DFields) offer more streamlined workflows for prescription map generation.

Facilities needed:

- Access to agricultural land for flight exercises. SLO County's vineyards, vegetable operations, and Cal Poly's agricultural facilities are potential sites.
- Computer lab access (shared with AERO 110 if processing software overlaps).

Departmental collaboration:

- Agriculture department for AGPS 242 alignment and access to agricultural sites. Agriculture faculty input on which crop systems and management practices are most relevant to the Central Coast region.
- Geography/Geology department if students also pursue GEOL 234 (Remote Sensing) as an elective.

Instructor development:

- Hands-on experience with multispectral data collection and interpretation in agricultural settings. This could be developed through partnerships with local vineyards or agricultural technology companies before the course launches.
- Understanding of precision agriculture workflows, including prescription map generation and integration with farm management software.

Open Questions

1. Course number: Where does this fit in the AERO numbering scheme?
2. Prerequisites: Should AERO 110 (Mapping) be a prerequisite, an advisory, or neither? Requiring it creates a longer path to the agriculture certificate. Making it an advisory signals that mapping background is valuable without creating a barrier. Borrowing content from AERO 110 into this course eliminates the dependency entirely, which is the current proposal.
3. Agriculture department buy-in: What is the appetite in the Agriculture department for collaboration? Do they see drone technology as relevant to their students?
4. Agricultural field access: Can we secure access to working agricultural operations for flight exercises? Cal Poly's research farms, local vineyards, or SLO County agricultural operations are all possibilities.

5. Regional market: Is the Central Coast agricultural market large enough to justify this course on local employment alone, or is this preparing students for statewide opportunities (Central Valley, Salinas Valley)?

AERO IXX: Agricultural Spray Operations

Purpose

Agricultural spraying represents the highest-revenue segment of agricultural drone work. This course covers spray drone operation, calibration and nozzle selection, drift management, chemical handling safety, and the regulatory stack required for legal operation in California. Graduates would be positioned for employment with agricultural spraying service providers, farm operations, and independent contract spraying.

This course is the most complex to bring to offering. The equipment is expensive, the regulatory requirements are extensive, and the instructor would need to develop expertise in an area outside current experience. It is included here because the market opportunity is real and growing – contract spraying rates of \$10–\$30 per acre translate to significant earning potential, and the Salinas Valley has recently seen commercial drone spraying operations launch. Whether this course is offered as part of the certificate program or as a standalone continuing education offering is an open question.

What It Takes to Offer

Equipment needed:

- Commercial spray drone such as the DJI Agras T40/T50 or Hyllo AG272 (\$20,000–\$30,000). These aircraft exceed 55 lbs loaded, which triggers additional FAA requirements.
- Chemical handling equipment and PPE.
- Calibration equipment for nozzle testing and flow rate verification.
- Field support vehicle for transporting equipment and chemicals.

The equipment investment is the highest of any proposed course by a wide margin.

Certifications and regulatory alignment: Students completing this course would need to pursue a stack of certifications beyond Part 107:

- FAA Part 137: Agricultural Aircraft Operator Certificate (the same certification required for manned crop dusters). Application approval takes 2–4 months.
- 44807 Exemption: Required for drones over 55 lbs. Must be included in the Part 137 exemption request.
- California QAC/QAL License: State pesticide applicator certification issued by the California Department of Pesticide Regulation. Requires passing exams on laws, regulations, and pest control principles.
- County Registration: Annual registration with each county of operation for legal pesticide application.

The course would need to align with DPR requirements to ensure students are prepared for the state licensing exams.

Facilities needed:

- Agricultural land for spray exercises. Operations would need to occur on permitted sites with appropriate buffer zones and drift management.
- Chemical storage meeting regulatory requirements.
- Wash-down and decontamination area.

Departmental collaboration:

- Agriculture department for pesticide application standards, safety protocols, and site access.
- Coordination with California DPR for applicator training alignment.

Instructor development:

- Part 137 certification and practical spray drone operation experience.
- QAC/QAL licensing or equivalent knowledge of California pesticide regulation.
- This represents the largest instructor development gap of any proposed course.

Open Questions

1. Offering format: Should this be a credit course within the certificate program, or a standalone continuing education offering for operators who already hold Part 107? The regulatory complexity and equipment cost may be better suited to a focused continuing education format.
2. Equipment funding: The spray drone alone costs \$20,000–\$30,000. Is this justifiable given the potential enrollment, or would a partnership with an agricultural operator who already owns the equipment be more practical?
3. Liability: Spray operations near crops and people carry different liability considerations than mapping or inspection flights. What are the insurance implications for the college?
4. Timing: The core program (AERO 100, 110, 120) needs to be established and running before this course makes sense. What enrollment and completion benchmarks should trigger development of this offering?
5. Industry partnerships: Are there agricultural spraying companies in the region willing to provide equipment access, guest instruction, or supervised field experience? This could reduce the upfront investment substantially.

Certificates of Specialization

Certificates of Specialization are Cuesta's smallest credential, typically consisting of two to four courses totaling 3 to 12 units. The college currently offers 23 of these certificates, with a median of 6.0 units. They target a single employable skill and are designed for quick completion. Four specializations are proposed for the UAS

program, each combining the AERO foundation with courses from partner departments.

UAS Mapping

Course	Title	Units	Department
AERO 100	Basic UAV Operator	3.0	Aerospace
AERO 110	Drone Mapping and Photogrammetry	2.0	Aerospace
GEOG 230	Introduction to Geographic Information Systems	3.0	Geology/Geography
	Total	8.0	

This specialization targets the largest commercial drone employment sector: mapping, surveying, and photogrammetry. Pairing AERO 110 with GEOG 230 gives students both the flight and processing skills to produce professional map products and the GIS foundation to work with the spatial data those products generate. GEOG 230 already exists and is the gateway course for Cuesta's 12-unit GIS Certificate of Achievement, so students in this track also begin progress toward that credential.

The collaboration with the Geology/Geography department is natural. Drone-collected data is increasingly central to GIS workflows, and Geography students benefit from understanding how that data is acquired. GEOG 230 is currently listed as an advisory on the AERO 110 Course Outline Report.

Career pathways: Surveying and engineering firms, construction site monitoring, environmental consulting, county and city planning departments, agriculture technology companies.

UAS Inspection

Course	Title	Units	Department
AERO 100	Basic UAV Operator	3.0	Aerospace
AERO 120	Thermal Imaging and Applied Inspection	3.0	Aerospace
	Total	6.0	

This specialization targets the infrastructure inspection and public safety sectors, unified by thermal imaging as the core technical skill. AERO 120 covers electrical infrastructure inspection, solar panel assessment, building envelope analysis, professional reporting workflows aligned with California utility regulations (General Orders 95 and 165), and public safety applications including search and rescue, fire assessment, and incident command integration.

PG&E alone operates over 80 drone pilots daily and completed more than 300,000 inspections in 2024, making utility inspection the single largest local employer of drone operators in our region. The job market research confirms that infrastructure

inspection is the dominant civilian employment pathway for thermal imaging skills. Public safety drone positions, by contrast, are typically filled by existing sworn personnel who add drone certification to their skillset rather than by external hires. AERO 120 serves both populations: students targeting civilian inspection careers and working first responders seeking professional development in thermal drone operations.

At 6.0 units and two courses, this certificate matches the institutional median for Certificates of Specialization and provides the fastest path to employment. Combined with the Part 107 certification from AERO 100 and the Level 1 thermography certification pathway in AERO 120, graduates hold two industry credentials that are standard requirements for inspection positions.

Career pathways: Utility inspection (PG&E, other IOUs), solar installation assessment, building inspection, insurance documentation, cell tower inspection, public safety agencies (for existing personnel adding drone skills).

UAS Agriculture

Agricultural drone work splits into two distinct disciplines that require different equipment, different certifications, and different knowledge bases. Rather than combining them into a single overloaded course, the program proposes two separate tracks. Both are Phase 2 offerings that should be developed in consultation with the Agriculture department.

Crop Monitoring Track

Course	Title	Units	Department
AERO 100	Basic UAV Operator	3.0	Aerospace
AERO 1XX	Multispectral Imaging and Crop Monitoring	2.0–3.0	Aerospace
AGPS 242	Introduction to Plant Science	3.0	Agriculture
	Total	8.0–9.0	

This track focuses on using multispectral drones to assess crop health, identify stress patterns, detect pest and irrigation issues, and generate prescription maps for variable-rate applications. The proposed AERO course would cover multispectral sensor operation, vegetation index calculation (NDVI and related indices), flight planning for agricultural missions, and data analysis workflows. It would borrow foundational mapping concepts from AERO 110 – flight planning, basic orthomosaic production, coordinate systems – but with a lighter treatment of ground control points and RTK positioning, focusing instead on the multispectral interpretation and agronomic analysis that distinguish this work. Color orthomosaic production would be covered as supporting context rather than as the primary deliverable.

Pairing the drone course with AGPS 242 (Introduction to Plant Science) gives students the agronomic foundation to interpret what the imagery is telling them.

Understanding plant structures, physiology, and crop management is essential context for translating vegetation indices into actionable recommendations. This keeps the certificate efficient – students do not need the full mapping course or the full remote sensing course to be productive in agricultural monitoring.

Equipment would require a multispectral-capable drone (DJI Mavic 3 Multispectral or equivalent, approximately \$6,500) in addition to the existing fleet. No additional FAA certifications beyond Part 107 are required.

Career pathways: Precision agriculture consulting, vineyard and orchard monitoring, farm management technology, agricultural research, crop insurance documentation.

Spray Operations Track

Course	Title	Units	Department
AERO 100	Basic UAV Operator	3.0	Aerospace
AERO 1XX	Agricultural Spray Operations	2.0–3.0	Aerospace
AGPS 242	Introduction to Plant Science	3.0	Agriculture
	Total	8.0–9.0	

This track addresses the highest-revenue segment of agricultural drone work: aerial application of herbicides, fungicides, insecticides, and fertilizers. The proposed AERO course would cover spray drone operation, calibration and nozzle selection, drift management, chemical handling safety, and the regulatory stack required for legal operation in California: FAA Part 137 (Agricultural Aircraft Operator Certificate), the 44807 exemption for aircraft over 55 lbs, and the California QAC/QAL pesticide applicator license.

This track requires substantially different equipment and carries higher costs than other specializations. Commercial spray drones (DJI Agras T40/T50, Hylío AG272) cost \$20,000–\$30,000, and the regulatory pathway is more complex. However, the earning potential is significant: contract spraying rates of \$10–\$30 per acre translate to peak-season revenues of several thousand dollars per day for operators with their own equipment. The Salinas Valley has recently seen commercial drone spraying operations launch, and vineyard spraying on the Central Coast represents a growing niche.

The spray operations course could also be structured as a continuing education offering for students who have already completed other AERO courses and want to add this specialization. Part 137 certification, in particular, may be better suited as a post-program add-on than as an embedded requirement, allowing students to complete a foundational certificate first and pursue spray certification as their career develops.

Career pathways: Agricultural spraying service providers, farm operations, vineyard management, agricultural cooperatives, independent contract spraying.

UAS Videography

Course	Title	Units	Department
AERO 100	Basic UAV Operator	3.0	Aerospace
ART 249A	Digital Photography I	3.0	Art
FTVE 225B	Single Camera TV Production and Editing	3.0	Film/TV/Electronic Media
	Total	9.0	

This specialization pairs drone flight skills with the visual composition and video production fundamentals that distinguish professional aerial videographers from hobbyists. ART 249A covers the technical and aesthetic foundations – exposure, composition, the elements of design, and camera technology in a digital context. FTVE 225B builds on those fundamentals with hands-on single-camera video production: scripting, shooting on location, portable lighting, audio, editing, and post-production. Together these courses give students both still image composition skills and motion video production capability alongside the drone operation skills from AERO 100.

Rather than developing new videography curriculum within the AERO prefix, this track leverages two existing courses from partner departments. The collaboration benefits all three programs: Art and FTVE students gain a compelling new tool for their production work, while AERO students gain skills that are otherwise absent from the drone curriculum. Both ART 249A and FTVE 225B are existing courses with no prerequisites beyond AERO 100, keeping the path to certification efficient.

Career pathways: Real estate media, event and wedding videography, tourism and destination marketing, documentary and news production, social media content creation, corporate video.

Certificate of Achievement: UAS Technology

The Certificate of Achievement is the program's comprehensive workforce credential, designed for students who want broad competency across multiple UAS application areas. Cuesta currently offers 63 Certificates of Achievement with a median of 19.5 units. The proposed structure falls within the institutional sweet spot of 15 to 20 units identified in the certificate landscape analysis.

Proposed Structure

Required Core (8.0 units)

Course	Title	Units
AERO 100	Basic UAV Operator	3.0
AERO 110	Drone Mapping and Photogrammetry	2.0
AERO 120	Thermal Imaging and Applied Inspection	3.0

Electives (select 9 units from the following)

Course	Title	Units
GEOG 230	Introduction to Geographic Information Systems	3.0
GEOL 231	Advanced GIS	3.0
GEOL 234	Remote Sensing	3.0
AGPS 242	Introduction to Plant Science	3.0
ART 249A	Digital Photography I	3.0
FTVE 225B	Single Camera TV Production and Editing	3.0
AERO 1XX	Multispectral Imaging and Crop Monitoring (when available)	3.0
AERO 1XX	Agricultural Spray Operations (when available)	3.0

Total: 17.0 units (8.0 core + 9.0 elective)

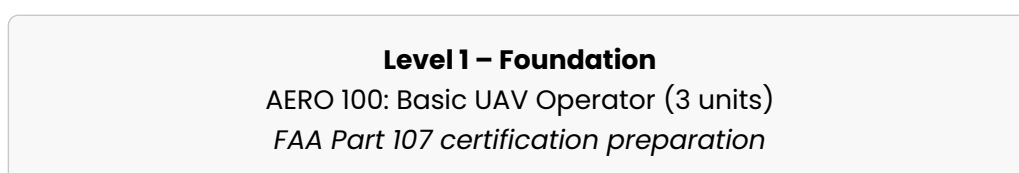
This structure ensures every graduate has both mapping and inspection competency from the three AERO courses, then allows specialization through the elective block. Students who have already earned a Certificate of Specialization will find that their coursework counts directly toward the Certificate of Achievement, creating a natural stacking pathway with no wasted units.

The elective list draws primarily from existing courses at the college, which means no new curriculum development is required in partner departments. Students are free to mix electives across disciplines or concentrate in a single area, depending on their career goals.

At 17.0 units, the certificate is comparable to Cloud Computing (16.0 units), Audio Technology I (16.0 units), Welding Technology (18.5 units), and Farm Management (18.0 units), all successful CTE certificates at Cuesta.

Stacking Model

The following diagram illustrates how credentials build on each other:



Level 2 – Certificate of Specialization (6–9 units)

AERO 100 + one applied course + one partner department course

Four tracks: Mapping, Inspection, Agriculture, Videography

Level 3 – Certificate of Achievement (17 units)

All three AERO courses + 9 units of electives

Comprehensive UAS Technology credential

This mirrors the Child Development model (Associate Teacher feeds into Teacher, which feeds into Master Teacher) and the Business Career Track model (common core feeds into concentration-specific certificates). Students enter at AERO 100, earn a specialization certificate with one additional semester of coursework, and can continue to the full Certificate of Achievement with one more semester. The entire pathway from entry to the comprehensive credential can be completed in three semesters, or two semesters for students who take a heavier course load.

Every course taken at the specialization level counts toward the Certificate of Achievement. A student who earns the UAS Mapping specialization (8 units) has already completed 8 of the 17 units required for the Certificate of Achievement. This zero-waste stacking approach encourages completion and reduces time to credential.

Cross-Enrollment Opportunities

A key advantage of this structure is that it creates enrollment pipelines between departments. The following existing courses serve double duty, appearing in both a UAS certificate and their home program's certificate or degree:

Course	UAS Certificate Role	Also Serves
GEOG 230	UAS Mapping specialization + C.A. elective	GIS Certificate of Achievement (12 units)
GEOL 231	C.A. elective	GIS Certificate of Achievement
GEOL 234	C.A. elective	GIS Certificate of Achievement
AGPS 242	UAS Agriculture specializations + C.A. elective	A.S.-T. Agriculture Plant Science
ART 249A	UAS Videography specialization	Photography Certificate of Achievement (17 units)

Course	UAS Certificate Role	Also Serves
	tion + C.A. elective	
FTVE 225B	UAS Videography specialization + C.A. elective	A.S.-T. Film, Television, and Electronic Media

This cross-pollination is standard practice at Cuesta. The certificate landscape analysis found 160 distinct courses appearing in more than one certificate program. By designing the UAS certificates to incorporate existing courses rather than creating parallel curriculum, the program contributes enrollment to partner departments while drawing students from those disciplines into AERO coursework.

Institutional Precedent

This proposal does not introduce a new structural pattern. It applies models the college already uses successfully.

Design Element	Existing Precedent
Common core branching into specialization tracks	Business Administration Career Track
Stackable credentials from small to large	Child Development (Associate Teacher to Site Supervisor)
Nested certificates where smaller feeds into larger	Aviation Mechanic (General to Airframe & Powerplant)
Cross-departmental course sharing	160 courses appear in multiple certificate programs
Certificates of Specialization at 6–9 units	23 existing certificates, median 6.0 units
Certificates of Achievement at 15–20 units	Institutional sweet spot for CTE credentials

The Aviation Mechanic program within the same department is the closest analog. AVMT 110 and AVMT 120 form the General certificate (12 units), which stacks into Airframe (30 units) and Powerplant (30 units), which combine into the full Airframe & Powerplant certificate (48 units). The UAS program applies this same nesting logic at a smaller scale appropriate to an emerging field.

Next Steps

This proposal outlines the structural framework. The following items need resolution before formal submission:

1. **Confirm course numbers and units.** AERO 110 unit count (2.0 vs. 3.0) is still under discussion. The agricultural courses need numbers and formal scope.
2. **Secure departmental buy-in.** Conversations with Geography, Agriculture, Art, and FTVE faculty will confirm the specific courses for each specialization and ensure those departments support including their courses in UAS certificates. The AERO 110 COR already identifies GEOG 230 as an advisory, so that relationship is partially established.
3. **Develop agricultural course proposals.** The two agricultural courses (crop monitoring and spray operations) need Course Outline Reports developed in consultation with Agriculture faculty. The crop monitoring course can borrow foundational content from AERO 110 while focusing on multispectral workflows. The spray operations course requires coordination with California DPR for applicator training alignment and a significant equipment investment decision.
4. **Draft program-level outcomes.** Each certificate needs 3 to 5 measurable outcomes. The institutional norm is to focus on specific competencies employers expect.
5. **Submit through CurriQunet.** Course Outline Reports for AERO 110 and AERO 120 are drafted. Certificate program proposals are a separate submission that references the approved courses.